

Cumulative Paper 1 — Number, Algebra, Ratio and Proportion

MTOH • NON-CALCULATOR • 60 MINUTES • 80 MARKS

NAME

DATE

SCORE / 80

- Answer every question in the space provided. Show your working — method marks are awarded even when the final answer is wrong.
- No calculator. Anything you cannot do in your head, do on paper.
- The mark for each question is shown in square brackets on the right. There are 80 marks in 60 minutes — about 45 seconds a mark, so do not sit on one question.
- Give every fraction and every ratio in its simplest form, and convert to a common unit before simplifying.
- Section 1 is recall, Section 2 is method, Section 3 is problems with no obvious first step. Work in order and do not spend more than five minutes stuck.
- Several questions are built so that a wrong method produces an answer that looks reasonable. Checking is part of the work.

SECTION 1 — THE MECHANICS

- 1.** Write 504 as a product of its prime factors. Give your answer in index notation. [2]

- 2.** Two numbers are 84 and 126.

- (a)** Find the highest common factor (HCF) of 84 and 126. [1]

- (b)** Find the lowest common multiple (LCM) of 84 and 126. [1]

One factor tree for each number serves both parts. Do not build it twice.

- 3.** Work out:

- (a)** $-12 \times 5 \div (-4)$ [1]

(b) $(-2)^3 \times (-3)^2$ [1]

4. Work out $\sqrt[3]{(-216)} + \sqrt{196}$. [2]

5. Simplify. Leave each answer in index form.

(a) $(3^5 \times 3^4) \div 3^6$ [1]

(b) $(4^3)^2 \times 4 \div 4^5$ [1]

6. Work out:

(a) $7.2 \div 0.1$ [1]

(b) $0.6 \times 0.01 \div 0.1$ [1]

Dividing by 0.1 makes a number larger. If your answer to (b) is smaller than 0.006 you have gone the wrong way.

7. Round 0.0498317

(a) to 3 decimal places, [1]

(b) to 3 significant figures. [1]

8. Write these decimals in ascending order. [2]

0.505 0.55 0.05 0.5 0.055

9. Write $\frac{7}{12}$ as a decimal. Use dot notation to show which digit recurs. [2]

10. Write down the single decimal multiplier for:

(a) an increase of 17% [1]

(b) a decrease of 6.5% [1]

11. Write the ratio 0.75 : 1.8 : 2.25 in its simplest whole-number form. [2]

SECTION 2 – METHOD

12. A taxi charges a fixed fee of ₹c plus ₹m for every kilometre travelled.

(a) Write a formula for the total cost, ₹T, of a journey of k kilometres. [1]

(b) The fixed fee is ₹43. A journey of 12 km costs ₹235. Work out the value of m. [2]

13. Expand and simplify:

(a) $5(3y - 2) - 2(4y - 7)$

[2]

(b) $x(x - 6)$

[1]

In (a), -2×-7 is plus 14. That one sign is where this question is usually lost.

14. Factorise fully:

(a) $12a + 18$

[1]

(b) $15x^2y - 25xy^2$

[2]

“Fully” means nothing is left inside the bracket that both terms still share.

15. Solve $(5x - 3)/4 = x + 2$. Show every step, then check your answer.

[3]

16. $N = 2^4 \times 3^5 \times 5^2$

(a) Explain how you can tell, without working out N, that N is not a square number.

[1]

(b) Find the smallest positive integer k for which Nk is a square number. [2]

Write a square number in prime factors and look at the indices. What do they all have in common?

17. Work out, showing your method:

(a) 3.6×0.45 [2]

(b) $0.126 \div 0.03$ [1]

18. Work out $5\frac{1}{6} - 2\frac{3}{4}$. Give your answer as a mixed number in its simplest form. [3]

One sixth minus three quarters is negative. Either borrow, or turn both into improper fractions first.

19. Work out, giving each answer in its simplest form:

(a) $6 \times 2\frac{5}{8}$ [2]

(b) $8 \div \frac{2}{5}$ [1]

In (b): dividing by a fraction smaller than 1 gives an answer bigger than 8.

- 20.** Write these fractions in descending order. You must show your working. [3]

$$\frac{5}{8} \quad \frac{7}{12} \quad \frac{11}{18} \quad \frac{3}{5}$$

Two of these four are very close together. Eyeballing them will not separate them.

- 21.** Amir, Bina and Cara share ₹9300. Amir's share : Bina's share = 3 : 4, and Bina's share : Cara's share = 6 : 5. Work out how much each person receives. [3]

The two ratios disagree about Bina. Scale them until they agree, then you have a three-part ratio.

- 22.** A laptop costs ₹8400. In a sale the price is reduced by 15%. Two months later the sale price is increased by 15%.

- (a)** Work out the final price of the laptop. [2]

- (b)** Find the overall percentage change from the original price. [1]

SECTION 3 – PROBLEMS

- 23.** A machine fills 1250 bottles in 25 minutes, working at a constant rate.

- (a)** Work out how many bottles it fills in 1 hour 12 minutes. [2]

- (b)** A second machine fills bottles at $\frac{3}{5}$ of the rate of the first. Both machines run together. Find how many minutes they take to fill 6400 bottles. [3]

Find what one minute is worth first. Everything else in this question comes off that number.

- 24.** A rectangle has length $(3x + 2)$ cm and width $(x - 1)$ cm. Its perimeter is 58 cm.

- (a)** Form an equation in x and solve it. [3]

- (b)** Hence find the area of the rectangle. [2]

Check your value of x by putting the length and width back into the perimeter before you do (b).

- 25.** A sports club has 448 members. This is after a 12% increase on last year's membership.

- (a)** Work out the number of members last year. [2]

- (b)** The 448 members are junior : senior in the ratio 5 : 3. Next year the number of juniors increases by 20% and the number of seniors decreases by 25%. Find the new ratio of juniors to seniors in its simplest form. [3]

In (a) the 448 is the new figure. Taking 12% off 448 is not the reverse of adding 12% to something else.

26. A sheet of card is 0.08 cm thick and has a mass of 4.6 g.

- (a)** Work out how many sheets are in a stack 2.4 cm tall. [2]

- (b)** Find the total mass of this stack in kilograms, correct to 2 significant figures. [3]

27. Rahul has handed in this homework. All three answers are wrong. For each one, say what he did wrong and write the correct answer.

- (a)** " $4.8 \div 0.1 = 0.48$ " [2]

- (b)** " $12 : 18 : 30$ in its simplest form is $6 : 9 : 15$ " [2]

(c) “ $0.7 \times 0.4 = 2.8$ ”

[1]

A mark for spotting the mistake, a mark for fixing it. Naming the rule he broke is worth more than just writing the right number.

End of paper. Go back to any answer that came out as a round, comfortable number — those are the ones a wrong method produces most often.